

Sushmita Mandloi

Ashta, Madhya Pradesh | +91 9244168599 | sushmitamandloi05@gmail.com | linkedin.com/in/sushmita-mandloi-6010112a8 | github.com/sushmita-mandloi

Engineering Highlights

Architected and deployed 4 end-to-end AI/ML projects — LLM-powered analysis agent, predictive maintenance system, demand forecasting pipeline, and customer analytics dashboard — using Python, XGBoost, LangChain, and Power BI
Reduced manual data-analysis time by 80% with InsightMate, an autonomous LLM agent integrating SQL, Python REPL, and data-visualization tools — deployed on Streamlit Cloud, serving 100+ live user queries

Education

Vellore Institute of Technology

2022 – 2027

Integrated M.Tech (B.Tech + M.Tech) in Computer Science & Engineering — CGPA: 8.08/10

– Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Statistical Analysis, Deep Learning, Natural Language Processing

Experience

Data Science Intern

Dec 2024 – Feb 2025

CodeSoft

Remote

- Delivered 5+ Python-based ML and data-analysis pipelines (Pandas, NumPy, Scikit-learn), improving downstream data-processing efficiency by 30%
- Executed end-to-end ETL workflows — preprocessing, feature engineering, and validation — across 10,000+ records, achieving 99% data accuracy and completeness
- Produced 3+ statistical predictive models with insights visualized in Matplotlib and Excel, informing data-driven decisions across 2 business teams

Projects

InsightMate – AI Data Analysis Agent | Python, LangChain, Groq LLaMA 3, Streamlit

GitHub

- Architected a dual-mode AI agent using the LangChain ReAct framework and Groq LLaMA 3.3-70B, integrating 4+ autonomous tools for data analysis including Python REPL, DuckDB SQL, Plotly, and PyMuPDF
- Reduced manual data-analysis time by ~80% by automating multi-step analytical workflows end-to-end via tool-chained LLM reasoning
- Engineered an LLM-powered resume parser using NLP extraction techniques, surfacing 10+ structured data fields at 90%+ accuracy
- Shipped a production-ready application on Streamlit Cloud with GitHub CI/CD integration, accumulating 100+ live user queries

Customer Shopping Behaviour Analysis | Python, SQL, Power BI, Pandas, NumPy, Matplotlib

- Automated an ETL pipeline in Python and SQL to clean and transform 5,000+ transactional records, surfacing purchasing trends and customer segmentation insights
- Engineered 15+ SQL analytical queries — aggregations, filtering, and grouping — quantifying revenue contribution and sales KPIs across 5+ business dimensions
- Designed an interactive Power BI dashboard with 10+ visualizations tracking revenue, segmentation, and subscription KPIs, improving reporting efficiency by 30%

Predictive Maintenance System | Python, XGBoost, Scikit-learn, Streamlit, Plotly

- Built a 2-stage predictive maintenance pipeline analyzing 3 sensor modalities (temperature, vibration, pressure) across 100 simulated machines, extracting 6 time-series features via 5-cycle rolling averages
- Trained an Isolation Forest for unsupervised anomaly detection and an XGBoost Regressor (100 estimators, max_depth=5) for Remaining Useful Life prediction, flagging failures 30+ cycles in advance on a 20% held-out test set
- Launched an interactive Streamlit dashboard with real-time Plotly monitoring and an ROI estimator, projecting an 80% cost reduction (\$40,000 saved per incident) compared to reactive unplanned downtime

Technical Skills

Languages: Python, SQL

Data Science & ML: Pandas, NumPy, Scikit-learn, XGBoost, Statistical Analysis, Feature Engineering, ETL Pipelines, Machine Learning, EDA

LLM & AI: LangChain, Groq LLaMA 3, Prompt Engineering, NLP, Sentiment Analysis, Text Classification

BI & Visualization: Power BI, Tableau, Matplotlib, Streamlit, Excel

Developer Tools: Git, GitHub, MLflow, VS Code, Jupyter Notebook

Databases: PostgreSQL, SQLite, DuckDB

Certifications & Achievements

SQL (Advanced) — HackerRank (2026); HackerRank 5-Star Badge in SQL

Quantium Data Analytics Job Simulation — Forage (2026)

Applied Machine Learning in Python — University of Michigan via Coursera (2023)

Leadership & Volunteering

NSS Club: Coordinated 3+ social awareness drives and community service initiatives across campus, managing logistics for 50+ student volunteers

Kanchan Education and Welfare Society: Conducted field assessments across rural schools, contributing findings that directly informed NGO program planning and resource allocation